

Engineering Project Report

LED ON/OFF Fading Dimmer Circuit Using NE555 Timer IC and
BC557 PNP Transistor

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Abstract

This report details the operational principles, component selections, mathematical modeling, and experimental assembly of an LED ON and OFF / Fading Dimmer Circuit. The design utilizes an NE555 precision timer configured as a low-frequency astable multivibrator, generating a continuous pulse train. The output drives a BC557 PNP Bipolar Junction Transistor (BJT) configured as a high-side electronic switch controlling a 6-LED array powered by a 12 V DC supply. Incorporating an RC smoothing network at the base of the BC557 converts sharp rectangular switching into smooth exponential fading transitions. Complete pinout specifications, component calculation methods, build instructions, and performance verification are included.

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1 Introduction & System Overview

Light-Emitting Diodes (LEDs) are widely used in low-power lighting, signal indication, and decorative effects. Creating soft pulsing or flashing visual effects enhances user experience without needing programmable microcontrollers.

This project uses an NE555 timer IC to set pulse rates and a BC557 PNP transistor to handle current switching across an array of LEDs.

2 Bill of Materials

Table 1: Bill of Materials for 555 + BC557 LED Driver Circuit

S.No.	Component	Qty	Part Number	Circuit Function
1	Timer IC	1	NE555N	Astable pulse generator
2	High-Side Transistor	1	BC557 (PNP BJT)	High-side current switch for LED array
3	LED Resistors	6	1 k Ω , 1/4W	Individual current limiters for LEDs
4	Base Limit Resistor	1	56 k Ω , 1/4W	Base drive current limiter
5	Biasing Resistor	1	47 k Ω , 1/4W	RC fading network resistor
6	Timing Resistors	2	100 k Ω , 200 k Ω	Astable timing feedback resistors
7	Timing Capacitors	2	10 μ F, 470 μ F	Astable timing (10 μ F) and fading (470 μ F)
8	Output Indicators	6	5mm LEDs	Visual array output
9	Power Supply	1	12 V DC Adapter	Main circuit DC supply

3 Transistor Switching Dynamics & Operating Principles

3.1 Astable Multivibrator Oscillation

The NE555 operates in astable mode. Capacitor C_1 (10 μ F) charges through $(R_A + R_B) = (200 \text{ k}\Omega + 100 \text{ k}\Omega)$ up to $\frac{2}{3}V_{CC}$ and discharges through $R_B = 100 \text{ k}\Omega$ to Pin 7 down to $\frac{1}{3}V_{CC}$. This outputs a continuous square wave on Pin 3.

3.2 BC557 PNP High-Side Switching Modes

A PNP transistor conducts when its Base voltage is driven lower than its Emitter voltage ($V_B < V_E = +12 \text{ V}$):

- 555 Output Pin 3 is LOW (0 V):** V_B drops below V_E , forward-biasing the Base-Emitter junction ($V_{BE} \approx -0.7 \text{ V}$). The BC557 turns ****ON**** (saturates), passing +12 V to the Collector and illuminating the LED array.
- 555 Output Pin 3 is HIGH (+12 V):** V_B equals V_E ($V_{BE} \approx 0 \text{ V}$). The BC557 turns ****OFF**** (cutoff), cutting off Collector current and extinguishing the LED array.

3.3 RC Fading / Dimming Transition

Connecting a 470 μ F capacitor in parallel with the 47 k Ω base network creates an RC time constant ($\tau = R \cdot C = 47 \text{ k}\Omega \times 470 \text{ }\mu\text{F} \approx 22.09 \text{ seconds}$). As base current charges and discharges exponentially, the LED brightness smoothly fades IN and OUT instead of flashing abruptly.

4 Assembly & Construction Guide

1. Place NE555 IC and BC557 transistor on the breadboard.
2. Connect 555 Pin 8 (V_{CC}) and BC557 Emitter to +12 V rail. Connect 555 Pin 1 to GND rail.
3. Connect 200 k Ω resistor from +12 V to 555 Pin 7. Connect 100 k Ω resistor between Pin 7 and Pin 6. Jumper Pin 6 to Pin 2.
4. Connect 10 μ F capacitor from Pin 2/6 to GND.
5. Connect 56 k Ω resistor from 555 Pin 3 to BC557 Base. Connect 470 μ F capacitor across the base network to +12 V.
6. Connect BC557 Collector to 6 parallel LED branches (each with a 1 k Ω resistor in series). Connect LED cathodes to GND.
7. Power with 12 V DC to observe smooth LED ON/OFF dimming behavior.

5 Conclusion

The NE555 + BC557 LED ON/OFF Dimmer Circuit cleanly illustrates PNP high-side switching and RC exponential fading dynamics, offering a simple analog visual effect generator.